Computer Project 2: Nonlinear MHD Plasma Dynamics, Toroidal Equilibria, Linear Stability

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This paper explores two different plasma confinement devices with three different numerical methods that can be used to characterize the behavior of each system. First, a three-dimensional ideal Magnetohydrodynamics simulation is used to study the sausage and kink instability of a z-pinch. Through adequate perturbation seeding the two modes can be excited and visualized with the expected dynamic. Next, the focus is changed to axisymmetric toroidal confinement devices. In a first step, the Grad-Shafranov equation is used to find the equilibrium flux profiles for different parameters of a specific solution of the problem. For the studied problem there exists a precise analytical solution which is used as a comparison for the numerical simulation. The two solutions match with great detail. Lastly, a linear stability analysis of a spheromak's tilt mode is performed to study the growth rate and find the elongation threshold for stability. The simulation introduced a numerical instability that could not be tracked down, so the elongation threshold from literature could not be reproduced.

INTRODUCTION

NONLINEAR MHD PLASMA DYNAMICS

The complex dynamical behavior of plasmas can be approximated using the Magnetohydrodynamics (MHD) equations. MHD combines the Maxwell equations with fluid dynamics to form a magnetized fluid model. Analytical solutions to this set of equations exist only for a limited amount of special geometries. Thus, the use of numerical methods is essential for general solutions of more complex geometries or dynamics. In the following, a 3D simulation on a uniform cartesian grid will be implemented and used to study Z-Pinch instabilities.

For the simulation, the MHD equations are brought into the conservative form

$$\frac{\partial Q}{\partial t} + \nabla \cdot \mathbf{F}(Q) = 0, \tag{1}$$

where Q is the state vector containing all conserved quantities and F is the flux vector that describes the transport of those quantities through space. For the 3D case, the state is fully described by eight independent quantities. Charge density ρ, momentum $\rho \boldsymbol{v}$, magnetic field $\boldsymbol{B}$, and total energy E. The flux vector $\boldsymbol{F}$ has three components that describe the flux of those eight quantities for all three directions in space: x, y, and z. The explicit formulation can be found in the Jupyter Notebook in the appendix.

For the simulation, the following class structure is used in python:

- FDMGrid: Holds the simulation domain for the applied finite difference method (FDM). Contains all methods that are used to initialize the simulation domain.
- MHDEquations: Describes the three flux vectors in dependence of the state vector.
- LFSolver: Defines the 3D Lax-Friedrichs method step that is used to propagate the solution in time.
- MHDSimulator: Contains all required methods for running the simulation and visualizing the results.

A good way to test simulations for their physical accuracy is to study the dynamical behavior of the system after small perturbations. In the linear regime, MHD plasmas are characterized through magnetosonic and Alven waves that propagate at different velocities and in different directions. The analytical solution for those characteristic values can be determined using the system's flux jacobian. This results in the following relations:

$$c_S = \sqrt{\frac{\gamma p_0}{\rho_0}}, \tag{2}$$

$$v_A = \frac{B_0}{\sqrt{\mu_0 \rho_0}}, \tag{3}$$

$$c_{ms} = \sqrt{c_S^2 + v_A^2}, \tag{4}$$

$$v_\pm = \frac{1}{2}(c_{ms}^2 \pm \sqrt{c_{ms}^2} - 4 v_A^2 c_S^2 \cos(\theta)^2), \tag{5}$$

where c_S is the sound speed, v_A the Alven speed, c_{ms} the magnetosonic speed, and $v_\pm$ the speed of the fast $(+)$ and slow $(-)$ magnetosonic wave.

Numerically, the system is prepared in a uniform state ($\rho_0 = 1$, $\boldsymbol{v}_0 = 0$, $\boldsymbol{B} = 0.5\hat{z}$, $p_0 = 1$) and perturbed with a Gaussian in the center of the simulation domain. The plots (Figures 1 and 2) show that the simulation agrees with the analytical solutions for the wave speed of the fast magnetosonic and Alven wave after being excited by a pressure or velocity perturbation respectively. The fast magnetosonic waves travels in all directions, while the Alven wave only propagates along the direction of the initial magnetic field vector ($\hat{z}$).

The power of the numerical solution becomes especially apparent when crossing into the non-linear regime.

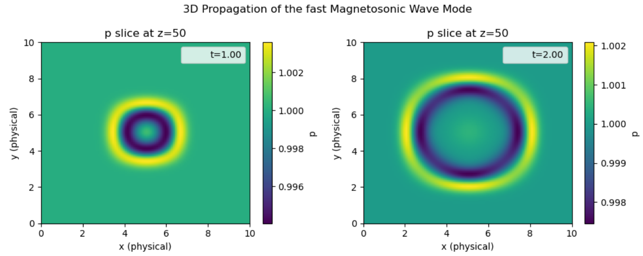

Figure 1. For this plot, the simulation domain was perturbed in the center by a small Gaussian pressure profile. This launches a fast magnetosonic wave travelling with approximately 1.4 units/s. This agrees with the analytical magnetosonic speed of 1.39 units/s.

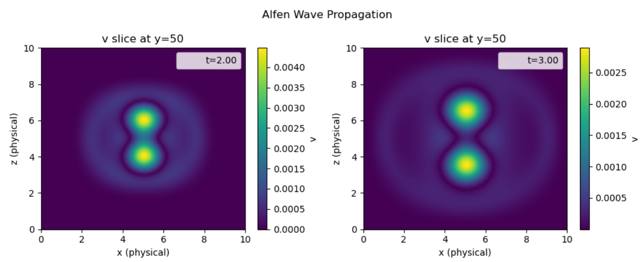

Figure 2. For this plot, the simulation domain was perturbed in the center by a small Gaussian velocity distribution. This perturbation travels in the direction of the initial magnetic field ($\hat{z}$) with a speed of approximately 0.5 units/s, perfectly agreeing with the analytical value of the Alven speed.

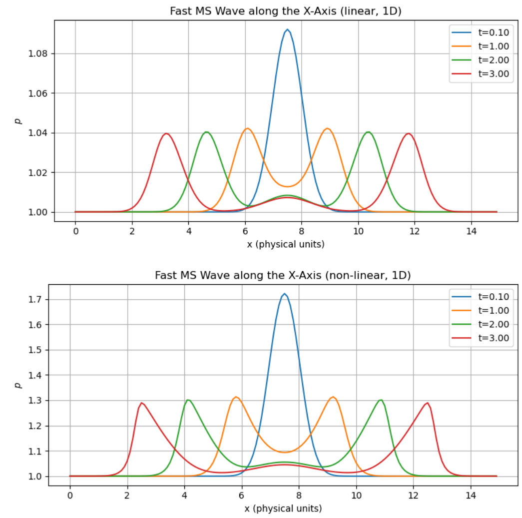

Figure 3. Comparing the wave propagation for the linear and non-linear regime, by changing the amplitude of the perturbation from 0.1 (top) to 0.8 (bottom). For the latter case, the peaks don't sustain their shape but get steeper towards their leading edge.

This means for perturbations that are so big that the linear approximation of the analytical solution doesn't hold anymore. While it is hard to describe this region analytically, nothing changes in complexity for the numerical solution compared to the linear case. Figure 3 shows that the wave shape changes between the two regimes, leading to different dynamical characteristics. This shows how the numerical solution enables the study of complex non-linear systems.

After the initial testing of the dynamical behavior of the simulation domain showed that the system is behaving as expected, the code will now be used to simulate the two dominating instabilities of a Z-Pinch. The Z-Pinch is a fusion device that uses an axial current and its corresponding azimuthal magnetic field to contain and compress a plasma. In its simplest form, the structure can be modelled inside a cube with solid conducting walls at the boundaries of the x and y direction and a periodic boundary condition in the z direction. At the center of this cubicle, parallel to the z-axis, a parabolic current profile with a radius of quarter a side length is initialized in a stable configuration.

From the current profile

$$J_z(r) = J_0 \cdot (1 - (r/a)^2), \tag{6}$$

results the magnetic field and pressure profile

$$B_\theta(r) = \frac{r}{2}\left(1 - \frac{1}{2}\left(\frac{r}{a}\right)^2\right) \tag{7}$$

$$p(r) = p(0) - J_0^2\left[\frac{r}{4} - \frac{3r^4}{16a^2} + \frac{r^6}{24a^4}\right]. \tag{8}$$

The simulation domain is initialized according to the steady-state above. For this analysis the following values are used: $J_0 = 1.0$, $B_{0,z} = 1.0$, and $p(0) = 1.0$. With these initial conditions an important metric for the Z-Pinch is set; the ratio between the stabilizing axial field and the azimuthal field at the pinch radius $B_z/B_a = 1/1.25 < 1$. As a first test, the pinch is evolved in time without any artificial perturbation. This shows the pinch naturally falling into an $m = 4$ instability that is caused by the discretization into a uniform cartesian grid (see Figure 4). To examine the dominant sausage ($m = 0$) and kink ($m = 1$) instabilities, an initial seeding is required that steers the system into the right direction.

Let's first consider the sausage instability, which is characterized by a partial reduction of the cross-section of the axial current. This is enforced by adding a perturbation that adds some momentum towards the axis of the pinch in the middle of the z-axis and some outward momentum at the edge of the domain. This small inequality grows with the unstable $m = 0$ mode leading to the characteristic splitting that looks like twisting two sausages apart visible in Figure 8.

The kink instability is marked by a deviation of the current flow away from the center of the axis to one side. It can be seeded by adding some radial and azimuthal momentum that are both dependent on the angle θ in

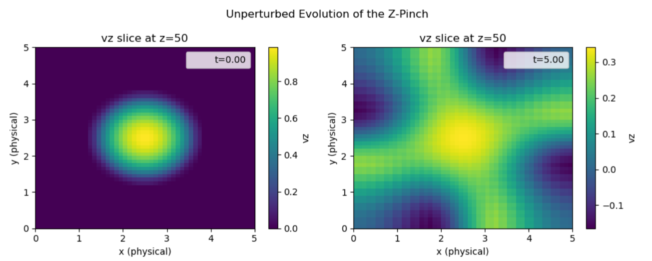

Figure 4. For this plot the simulation domain was initialized with the screw pinch equilibrium and then evolved over time. After 5 s the axial current split up into the center part plus four additional axes towards the edge of the simulation domain. This $m = 4$ is an artifact of the cartesian discretization and not the physically dominant instability.

the xy-plane. This creates the characteristic inequality that shows the axial current moving to one side, away from the center axis as seen in Figure 10. This instability can be mitigated by increasing the axial magnetic field, to get the above-mentioned ratio above 1.

Using the developed class structure one can now study any dynamical behavior that can be described by the ideal MHD model. Especially, combining the numerical simulation with experimental data can lead to a better understanding of the physical processes that characterize the individual system. Moreover, the simulation can be used to test how experimental conditions might influence the properties of the plasma, which can then be used to optimize the experiment.

TOROIDAL EQUILIBRIA

For the study of equilibrium states, a full dynamical 3D modelling of the plasma system is unnecessary. Instead, using equilibrium assumptions, the system can be collapsed into a set of time-independent elliptic equations. One example of an equilibrium equation is the so-called Grad-Shafranov equation

$$\Delta^* \psi(r,z) = -\mu_0 r^2 p'(\psi) - F(\psi) F'(\psi), \tag{9a}$$

$$\Delta^* \equiv r \frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}\right) + \frac{\partial^2}{\partial z^2}, \tag{9b}$$

$$p'(\psi) \equiv \frac{dp}{d\psi}, \qquad F'(\psi) \equiv \frac{dF}{d\psi}, \tag{9c}$$

that can be used to describe axisymmetric toroidal plasmas. Here ψ is the flux function that is constant along any magnetic field line, μ_0 the magnetic permeability, p the pressure, and $F = rB_\theta$ (radius times toroidal magnetic field). Given a pressure and magnetic field profile, as well as a complete set of boundary conditions, this equation will give the resulting equilibrium state of the system.

This equation can be derived from reducing the following equations:

- Gauss's law of magnetism: $\nabla \cdot \mathbf{B} = 0$
- Ampère's law: $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$
- MHD force balance: $\mathbf{J} \times \mathbf{B} = \nabla p$

The profiles for p and F are often approximated using power series expansions. For the purpose of this report, the two quantities will be fixed so that $p' = -a$ and $FF' = -br_0^2$. This is handy because there exists an analytical solution for this special case

$$\psi(r,z) = \frac{1}{2}\left(b\,r_0^2 + c\,r^2\right) z^2 + \frac{1}{8}(a-c)\,(r^2 - r_0^2)^2, \tag{10}$$

that can be used as a comparison for the generated numerical solution. Here c is a constant which can be determined by further specifying the equilibrium. Plugging this expression for ψ into the Grad-Shafranov equation doesn't yield a constraint on c showing that it is a purely geometrical parameter. For a toroidal equilibrium we expect the flux to change isotropic around the space very close to the magnetic axis. We can enforce this condition by replacing the radial coordinate by $x = r - r_0$ that gives the displacement away from the magnetic axis at r_0. After expanding the term one gets

$$\begin{aligned}\psi = {} & \frac{1}{2}(b+c)r_0^2 z^2 + \frac{1}{2}cx^2z^2 + cr_0xz^2 \\ & + \frac{1}{8}(a-c)4x^2r_0^2 + 4x^3r_0 + x^4.\end{aligned} \tag{11}$$

The flux very close around the magnetic axis at $x = z = 0$ is sufficiently described by only keeping the lowest order (quadratic) terms, leading to

$$\psi \approx \frac{1}{2}r_0^2[(a-c)x^2 + (b+c)z^2]. \tag{12}$$

Requiring the flux to be isotropic, this relation gives $a - c = b + c$, or more specific the desired constraint $c = (a-b)/2$.

For the numerical simulation, equation 9a needs to be discretized. This is done by first expanding the Δ^* operator and then approximating with second order centered differences:

$$\Delta^*\psi = \frac{\partial^2\psi}{\partial r^2} - \frac{1}{r}\frac{\partial\psi}{\partial r} + \frac{\partial^2\psi}{\partial z^2}, \tag{13}$$

$$\begin{aligned}(\Delta^*\psi)_{i,j} = {} & \frac{\psi_{i+1,j} - 2\psi_{i,j} + \psi_{i-1,j}}{(\Delta r)^2} \\ & - \frac{1}{r_i}\frac{\psi_{i+1,j} - \psi_{i-1,j}}{2\,\Delta r} \\ & + \frac{\psi_{i,j+1} - 2\psi_{i,j} + \psi_{i,j-1}}{(\Delta z)^2}.\end{aligned} \tag{14}$$

Here the index i refers to the radial and j to the axial dimension, while Δr and Δz are the grid spacing in the respective dimensions. This $\Delta^*\psi$ can also be written as a matrix expression A$\boldsymbol{x}$, when stacking the two dimensions in a one dimensional vector with entries $k = i + jN_r$ (N_r is the number of grid points in the r dimension).

Since the right-hand side of equation 9a is independent of ψ it can be combined into the source vector

$$\boldsymbol{b} = \mu_0 r_i^2 a + b r_0^2. \tag{15}$$

This vector is constant along z and varies quadratically along r.

Putting the left and right-hand side together, the linear system $\mathrm{A}\boldsymbol{x} = \boldsymbol{b}$ is created that needs to be solved for $\boldsymbol{x}$ to get the flux across the domain. This can be done by inverting A and calculating $\boldsymbol{x} = \mathrm{A}^{-1}\boldsymbol{b}$. Another option, with a possibly higher computational efficiency, is to use an iterative formula. This makes sense here since most of the matrix entries are 0. A simple method is to divide A into its diagonal matrix D and the off diagonal matrix O: A=D+O. Inserting this in to the linear system and rearranging leads to

$$\boldsymbol{x} = \mathrm{D}^{-1}(\boldsymbol{b} - \mathrm{O}\boldsymbol{x}). \tag{16}$$

Since O=A−D, this can be rearranged to

$$\boldsymbol{x}^{n+1} = \boldsymbol{x}^n + \mathrm{D}^{-1}(\boldsymbol{b} - \mathrm{A}\boldsymbol{x}^n), \tag{17}$$

where it was additionally assumed that the function of $\boldsymbol{x}$ will converge to the fixed point $\boldsymbol{x}^*$ that solves the linear system. With this equation, the solution of the linear system can be refined iteratively, leading to the final equilibrium field.

For a proper comparison between the numerical and analytical solution, we need to enforce equal boundary conditions for both cases. Here, this is implemented by using the values of the analytical solution for the source term at the boundaries of the numerical solution.

Figure 5 and 6 show the analytical and numerical solution respectively for the following relations of the parameters a and b: $b/a = 0, -1/7, -7, 1$. Comparing the two Figures shows a great match between the analytical and numerical solution. The only difference is visible very close around the magnetic axis, where the contours are visualized by black dashed lines. These lines were added to visualize the flux at the interior of the plasma and are not spaced according to some gradient. Since the values of the contours are very close around zero, the slight mismatch due to numerical accuracy is visible.

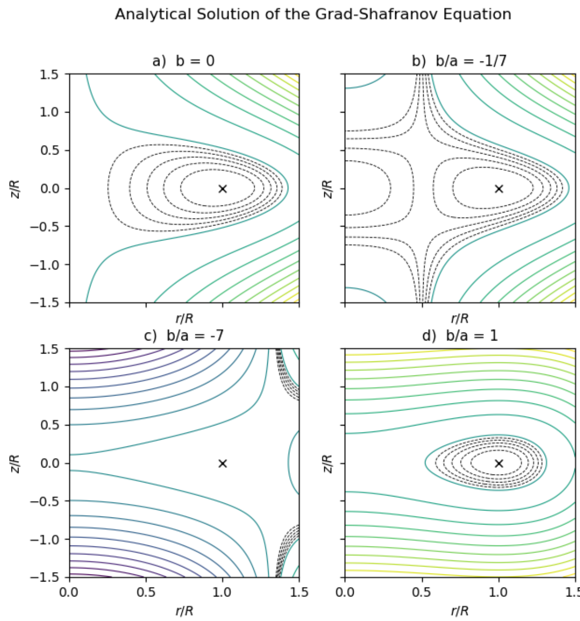

Figure 5. On this graphic, the flux field of the analytical solution is portrayed for four different values for b/a. Most plots show convincing equilibria with concentric rings around the magnetic axis. For case c) this is also true, but the author was unable to visualize the contours properly.

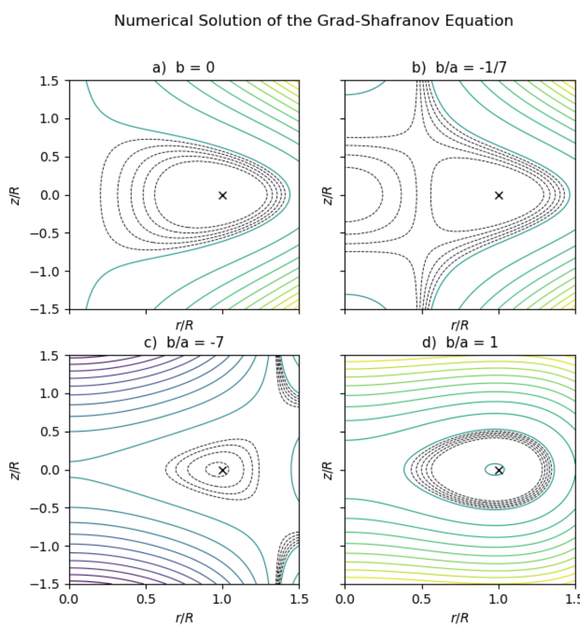

Figure 6. This Figure shows the flux field of the numerical solution for four different values for b/a. All plots show equilibria, characterized by concentric flux rings around the magnetic axis. Due to the uneven numerical simulation that is only accurate to second order, the algorithm that is employed for the contour visualization is also successful for case c).

LINEAR STABILITY ANALYSIS

Important for the study of equilibria, like the ones found in the previous section, is their stability towards perturbations. Is the equilibrium stable and counteracts any perturbations or are there certain modes that grow indefinitely? A powerful tool to answer this question is the Linear Stability Analysis. Its idea is to study small perturbations to an initial steady state so that second order perturbation terms can be neglected. Applying this to the existing set of MHD equations (the ones we studied in the first section), the new collection of linearized MHD equations can be obtained (here, we additionally assume a static pressure-less equilibrium, so $\boldsymbol{v}_0 = p_0 = 0$):

$$\frac{\partial \mathbf{B}_1}{\partial t} = \nabla \times \big(\mathbf{v}_1 \times \mathbf{B}_0\big), \tag{18a}$$

$$\rho_0 \frac{\partial \mathbf{v}_1}{\partial t} = -\nabla p_1 + \frac{1}{\mu_0}\Big[(\nabla \times \mathbf{B}_1) \times \mathbf{B}_0 + (\nabla \times \mathbf{B}_0) \times \mathbf{B}_1\Big]. \tag{18b}$$

For this stability analysis, the exploration of the axisymmetric toroidal equilibrium from last section will be continued, which means that the equations are solved on a slice of the rz-plane in cylindrical coordinates. In particular, the $m = 1$ tilt mode of a spheromak will be studied, so a formulation of the form

$$g_1(r, \theta, z, t) = \hat{g}(r, z, t)\, e^{i\theta} \tag{19}$$

is used to express all perturbed quantities g_1. This reduces the toroidal derivative to $\partial g_1/\partial\theta = i\, g_1$.

In order to discretize the partial differential equations, the ∇-terms are expressed as cartesian derivatives in cylindrical coordinates. These spatial first order derivatives are approximated using second order accurate finite difference schemes, while the time derivatives are implemented as forward Euler steps.

For the boundary conditions, conductive walls are enforced at z_{min}, z_{max}, and r_{max}, which means that the normal components of the velocity and magentic field are fixed at 0. At the axial boundary at $r = 0$, both the radial and azimuthal field components are set to 0, which ensures regularity.

The class structure in python that is used for this simulation is very similar to the one presented in the first section. One class holds the information about the grid and any methods that are needed for initialization or boundary condition treatment. Another class, defines the full set of PDEs and a method that advances the variables by one time-step dt. For the final step of the simulation, a class advances the solution with a variable time step, that keeps the solution numerically stable, until it reaches a given final time. During the time advance, this class collects the total energy of the system to later resolve its evolution in time. Finally, all analysis is performed with a separate class that can hold several simulation entries and create the plots that are used to characterize the tilt mode in the rest of this section.

To characterize the instability, a small velocity perturbation is initialized that will grow exponentially once the unstable mode is excited. The dynamical growth is then observed by plotting the total kinetic energy semi-logarithmically over time. The slope with which the energy grows in this plot is two times the growth factor γ (Figure 7). However, this is not the realistic growth factor, since this numerical solution is only accurate up to Δx^2. One approach to retrieve the real growth factor from this numerical simulation is to run the stability analysis for different grid sizes and plot the corresponding growth factors together on a plot with $(J_0/J)^2$ on the x-axis. Here, J_0 is the coarsest grid size, while J is the grid size corresponding to the plotted growth factor. While the point for the coarsest grid will lie at $(J_0/J)^2 = 1$, the finer grids will be closer and closer to the y-axis, evolving in linear fashion. Extrapolating this linear slope will lead to the intercept with the y-axis. There, at $(J_0/J)^2 = 0$, the grid size is infinite, corresponding to the real growth factor (On Figure 9, the general idea of the extrapolation is visualized. However, since the obtained physical data is bad - explanation below - the growth factor rises for finer grids, although the opposite is expected).

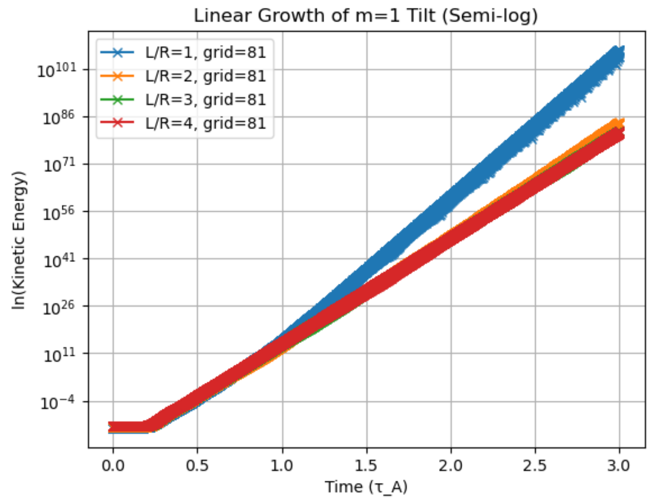

Figure 7. This semilogarithmic plot shows how the instability grows exponentially for all elongation values. At least for $L/A = 1$ one would expect to see no exponential growth since this elongation is well within the stability limit. However, since the simulation exhibits additional numerical instability, the plot shows nonphysical behavior.

A literature review shows that the stability is dependent on the elongation factor L/R which determines the size in the z and r direction respectively. The system is expected to be stable for elongations $\leq 1.65/1.68$. To validate this result, the author tried to determine the growth factor for different elongations in the search for stable and unstable systems. However, the implemented numerical simulation added some numerical error that made the determined growth factors nonphysical. The field plots for two different elongations (Figure 11) show that the instability doesn't have the desired shape characterized by two separated curls. Instead, the fields dominate on the axis, hinting to some sort of boundary value, or differencing problem that is mishandling the axis at $r = 0$. Although a serious effort was undertaken to eliminate this problem, the error source was not found making the determination of an elongation threshold impossible.

Appendix

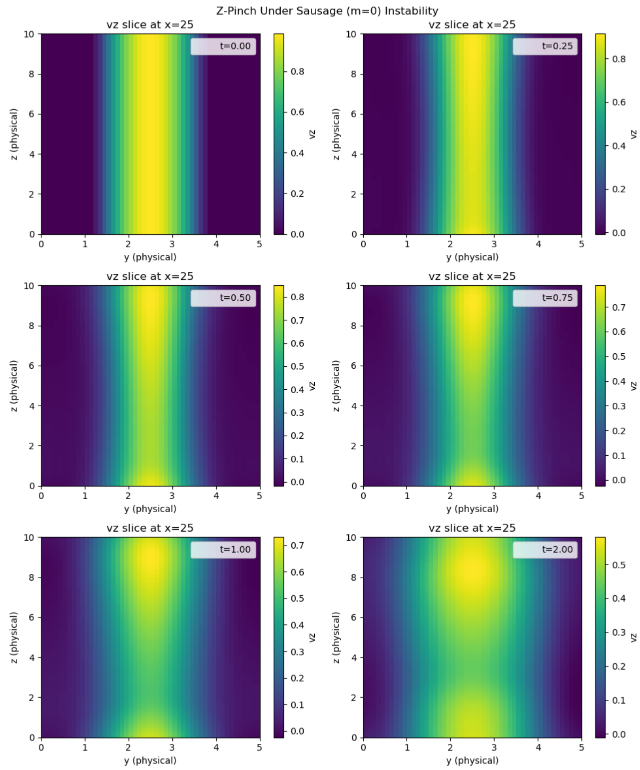

Figure 8. These plots show the time evolution of the $m = 0$ instability. The small initial perturbation grows quickly until the axial current is stopped completely.

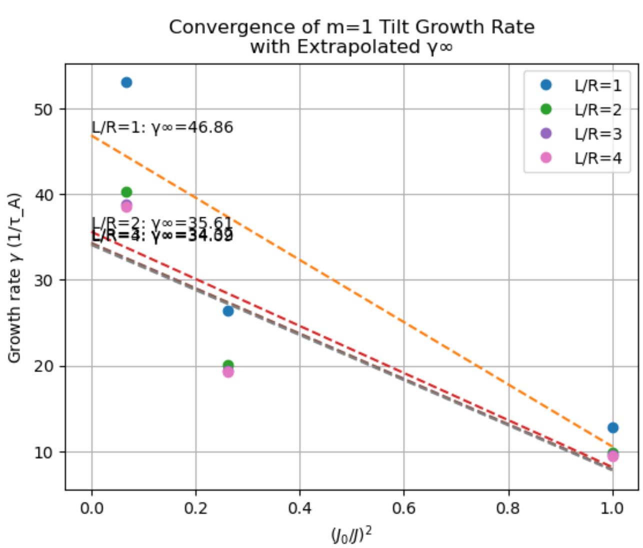

Figure 9. By performing the growth factor analysis for different grid sizes, one can extrapolate the linear slop to find the real growth factor at the intersection with the y-axis. Usually, a decreasing growth factor for finer grid sizes would be expected, so the numerical instability grows with grid size.

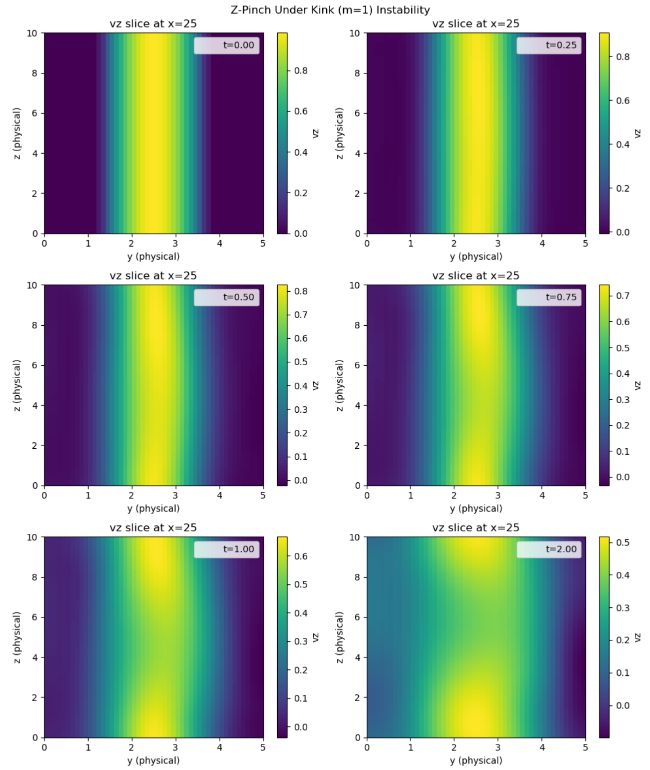

Figure 10. On this time evolution of the pinch, one can observe the $m = 1$ instability that steers the axial current to the side and eventually stops the flow completely.

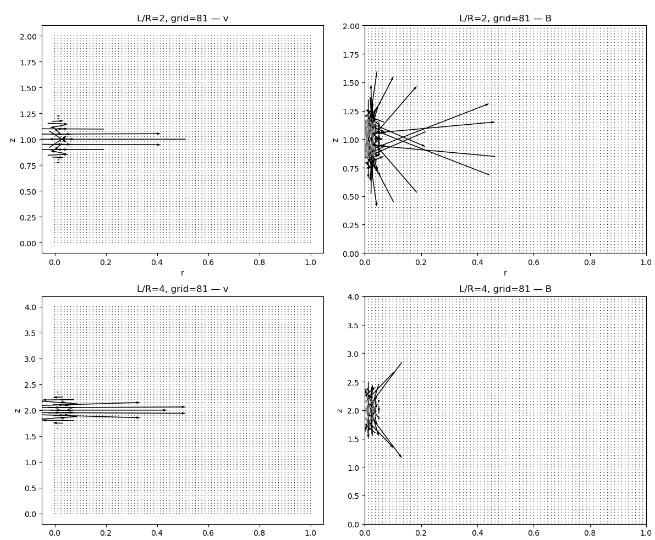

Figure 11. This field plot of the grown instability shows how the fields are maximal at the axis. Usually, they would be expected to draw two separated curl patterns above each other. The depicted behavior must be due to some error in the numerical implementation for the axis boundaries/derivatives.